Example 1.23 If $f(x) = 2x + 3, g(x) = 1 - 2x$ and $h(x) = 3x$. Prove that $f \circ (g \circ h) = (f \circ g) \circ h$

Solution $f(x) = 2x + 3$, $g(x) = 1 - 2x$, $h(x) = 3x$

$$\text{Now, } (f \circ g)(x) = f(g(x)) = f(1-2x) = 2(1-2x) + 3 = 5 - 4x$$

$$\text{Then, } (f \circ g) \circ h(x) = (f \circ g)(h(x)) = (f \circ g)(3x) = 5 - 4(3x) = 5 - 12x \quad \ldots.(1)$$

$$(g \circ h)(x) = g(h(x)) = g(3x) = 1 - 2(3x) = 1 - 6x$$

$$\Rightarrow f \circ (g \circ h)(x) = f(1-6x) = 2(1-6x) + 3 = 5 - 12x \quad \ldots.(2)$$

From (1) and (2), we get $(f \circ g) \circ h = f \circ (g \circ h)$

Example 1.24 Find x if $gff(x) = fgg(x)$, given $f(x) = 3x + 1$ and $g(x) = x + 3$.

Solution

$$gff(x) = g\,[f\{f(x)\}] \text{ (This means "}g \text{ of } f \text{ of } f \text{ of } x\text{")}$$

$$= g\,[\,f(3x+1)] = g\,[\,3(3x+1)+1] = g\,(9x+4)$$

$$g\,(9x+4) = [\,(9x+4) + 3] = 9x + 7$$

$$fgg(x) = f\,[g\,\{g\,(x)\}] \text{ (This means "}f \text{ of } g \text{ of } g \text{ of } x\text{")}$$

$$= f\,[\,g\,(x+3)] = f\,[\,(x+3) + 3] = f\,(x+6)$$

$$f\,(x+6) = [\,3(x+6) + 1\,] = 3x + 19$$

These two quantities being equal, we get $9x + 7 = 3x + 19$. Solving this equation we obtain $x = 2$.

Progress Check

State your answer for the following questions by selecting the correct option.

1. Composition of functions is commutative
   (a) Always true (b) Never true (c) Sometimes true
2. Composition of functions is associative
   (a) Always true (b) Never true (c) Sometimes true

Activity 4

Given that $h(x) = f \circ g(x)$, fill in the table for $h(x)$

x	$f(x)$
1	2
2	3
3	1
4	4

x	$g(x)$
1	2
2	4
3	3
4	1

x	$h(x)$
1	3
2	-
3	-
4	-

How to find $h(1)$?

$$h(x) = f \circ g(x)$$

$$h(1) = f \circ g(1)$$

$$= f(2) = 3$$

$$\therefore h(1) = 3$$

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